# Wake up your rover

**Name:** ______________________________

**Date:** ____________________ **Period:** ____________________

## Today I can...

Show a color to the sensor and explain the message that appears.

## Words to know

Sensor: A part that detects something, such as a color.

Program: The steps the robot follows.

Input / output: The color reading goes in. A screen message comes out.

**LOOK AT THE IDEA**

| WHITE | RED | OTHER |
|---|---|---|
| ↓ | ↓ | ↓ |
| READY | DELIVERY | CHECK |

## Gather your materials

- LEGO set #45522 and its official instructions
- Connected computer or tablet
- Red, white and blue paper: one 8 cm square of each
- Two equal-height firm supports, checked by the teacher

# Build a color-test station

Build with the power off. Tick each step when it is done.

1. Open LEGO's official instruction book. Find model C103 on pages 24-31. Build it with your teacher as a starting model, or use the wheeled base your teacher has already checked. The pictures in this course are project ideas, not exact LEGO assembly steps.

2. Ask the teacher to prepare a secure downward-facing color-sensor mount and clear attachment points for later lessons. Keep both wheel paths and all Stop controls clear. This is a change to the starting model and needs a real-kit test.

3. Set the robot on the two firm supports under fixed parts of its frame. Check that it cannot rock or slide. Both wheels must hang clear of the table. Do not support it by the sensor or an axle (a rod a wheel turns on).

4. Cut three paper squares, each 8 cm by 8 cm. Use red, white and blue. Put them beside the robot. Keep the same side of each paper facing the sensor.

Base reference: official LEGO C103 instructions, pages 24-31. Use the base your teacher has checked.

**Sketch the setup. Label the parts you will check.**

# Finish the setup

5. Connect only your team's motor and sensor in Coding Canvas. Point to the screen Stop control and the robot's power button. Ask the teacher to check the connection.

6. Slide white paper below the sensor without touching it. Adjust the paper position until the sensor reads white. Mark that place with tape so every card goes in the same spot.

7. Build the color-to-message program below. Keep all movement stopped. Test one card at a time, then put it back in the marked spot.

8. Record all three results. Change one screen message to your team's words and test again. Leave the sensor mount and base ready for lesson 2.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

**Show where your robot starts and how you will stop it.**

# Code it. Predict. Try it.

## Make the program

1. Start with all movement stopped.

2. Repeat 10 times: read the color. White shows READY; red shows DELIVERY; anything else shows CHECK. Wait 0.2 seconds before the next check.

3. Stop all movement and end. There are no driving steps in this program.

Coding Canvas is the app where you make the program. Use your teacher-checked settings before a floor run.

Before you run: what do you think will happen?

After one try: what actually happened?

End of class 1: save the code, stop and power off, label your parts, and keep these pages.

# Test it fairly

**Name:** ____________________

**Date:** ____________________ **Period:** ____________________

Change the paper color. Keep the light and the sensor’s position the same.

## A successful test

- ☐ The base is steady and both wheels stay still.
- ☐ Each color produces the message in the rule.
- ☐ Everyone can point to the sensor and the Stop controls.

## Record every result

| Card color | Expected message | Actual message | Wheels still? |
|---|---|---|---|
| White | | | |
| Red | | | |
| Other | | | |

Which result stands out?

____________________

# Change one thing. Try again.

The one thing we changed:

We kept these things the same:

| Card color | Expected message | Actual message | Wheels still? |
|---|---|---|---|
| White | | | |
| Red | | | |
| Other | | | |

How does the color make the screen message change?

Use a result: our change helped / did not help because...

Before leaving: save your code, stop and power off the robot, return parts, and hand in this module.

# Make a short drive

**Name:** ______________________________

**Date:** ____________________ **Period:** ____________________

## Today I can...

Make the robot stop after a short time. Measure how far it travels.

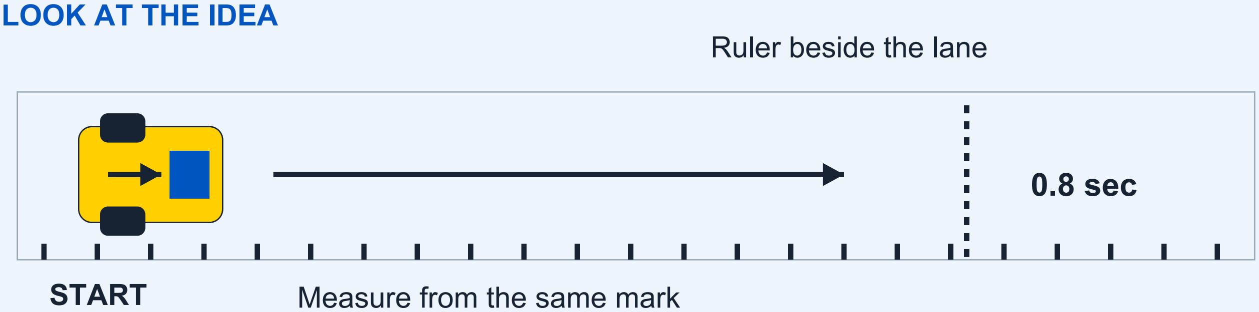
## Words to know

Motor: A part that makes something move.

Bounded: Set to stop after a certain time or number of steps.

Trial: One test run.

**LOOK AT THE IDEA**




## Gather your materials

- Checked wheeled base and connected device
- Ruler or measuring tape
- Removable floor tape
- Paper and pencil

# Build a measured driving lane

Build with the power off. Tick each step when it is done.

1. Choose a flat floor space with the teacher. Keep it away from stairs and table edges. Make the lane at least twice as wide as the robot.
2. Mark a start line across the lane. Put the ruler beside the lane, with zero at the start line. Keep the ruler out of the wheel path.
3. Add a small tape mark on the side of the robot, above its main wheel axle. Use this same mark to measure every run. An axle is the rod a wheel turns on.
4. With the power off, turn each wheel gently by hand. Remove anything rubbing a wheel. Check that the sensor mount stays clear of the floor.

**Sketch the setup. Label the parts you will check.**

# Finish the setup

5. Raise the wheels on firm supports. Try the short program below. If the wheels would send the robot backward, stop and ask the teacher to correct the motor direction.

6. Place the robot's measuring mark over the start line. Clear the lane. With the teacher's approval, run the program and wait until the wheels stop.

7. Measure from the start line to the same mark on the stopped robot. Write the distance in centimeters (cm). Return it to the start with the power off.

8. Repeat three times with the same code. Then ask the teacher to help test the Stop control during another short run. Save the lane and measuring mark.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

**Show where your robot starts and how you will stop it.**

# Code it. Predict. Try it.

## Make the program

1. Start with movement stopped. Set low speed, such as 20 percent.

2. Move forward for 0.8 seconds, then stop.

3. End the program. Do not use a forever-repeat block.

Coding Canvas is the app where you make the program. Use your teacher-checked settings before a floor run.

Before you run: what do you think will happen?

After one try: what actually happened?

End of class 1: save the code, stop and power off, label your parts, and keep these pages.

# Test it fairly

**Name:** ______________________

**Date:** ______________________ **Period:** ______________________

Change only one thing, such as speed. Keep the time, floor, load and start line the same.

## A successful test

- ☐ The robot stops by itself after each short drive.
- ☐ All three distances are recorded from the same mark.
- ☐ The teacher checks the screen Stop control and power-off method.

## Record every result

| Run | Speed setting | Distance (cm) | Stopped by itself? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

Which result stands out?

______________________

# Change one thing. Try again.

The one thing we changed:

We kept these things the same:

| Run | Speed setting | Distance (cm) | Stopped by itself? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

Why might three runs travel slightly different distances?

Use a result: our change helped / did not help because...

Before leaving: save your code, stop and power off the robot, return parts, and hand in this module.

# Build a parcel holder

**Name:** ____________________

**Date:** ____________________ **Period:** ____________________

## Today I can...

Build two versions of a holder and use your tests to choose one.

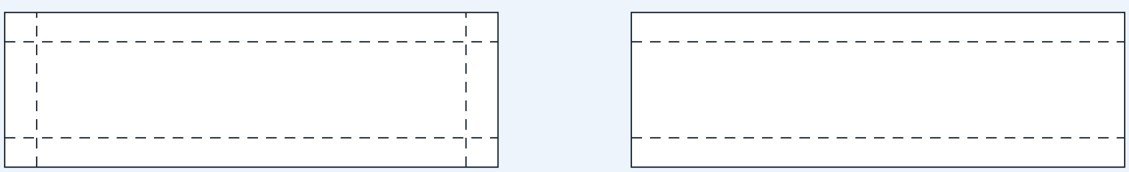
## Words to know

Cargo: The thing your robot carries.

Cradle: A holder that keeps the parcel in place.

Fair test: Change one thing and keep the rest the same.

**LOOK AT THE IDEA**



**A: four low walls**

**B: open short ends**

## Gather your materials

- Checked base with teacher-approved fixed mounting points
- Thin cardboard: two rectangles about 12 cm by 8 cm
- Paper parcel about 4 cm by 3 cm by 2 cm
- Ruler, pencil, scissors and removable tape

# Build two parcel holders

Build with the power off. Tick each step when it is done.

1. Fold a small paper parcel about 4 cm long, 3 cm wide and 2 cm high. Tape it closed. Use this same light parcel for every test. Do not add coins or other weights.
2. Measure the clear space your teacher chose on the base. A 12 by 8 cm card is a starting size. Make it smaller if it would hang over a wheel or cover a control.
3. For holder A, draw a line 1 cm inside each edge of one rectangle. Cut out the four corner squares. Fold the four sides up along the lines and tape the corners. You have made a low tray.
4. For holder B, use the second rectangle. Fold its two long edges up by 1 cm and keep both short ends open. Put two folded paper stops near the short ends to stop the parcel sliding out.

**Sketch the setup. Label the parts you will check.**

# Finish the setup

5. Turn the robot off. Ask the teacher to show the mounting points. Fix holder A to those fixed points with the approved reusable connectors or paper tape tabs. Do not tape over the motor, sensor or buttons.
6. Put the parcel in holder A. Raise the wheels and check for rubbing. Try the same short drive from lesson 2 three times and record whether the parcel stays on.
7. Turn the power off. Swap to holder B. Use the same parcel, code, floor and start line for three more runs.
8. Choose the better holder using your notes. Change only one feature if neither works well. Keep the selected holder for later lessons.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

**Show where your robot starts and how you will stop it.**

# Code it. Predict. Try it.

## Make the program

1. Use lesson 2’s checked program.

2. Low speed, 0.8 seconds forward, then stop and end.

3. Keep the same settings for both holders.

Coding Canvas is the app where you make the program. Use your teacher-checked settings before a floor run.

Before you run: what do you think will happen?

After one try: what actually happened?

End of class 1: save the code, stop and power off, label your parts, and keep these pages.

# Test it fairly

**Name:** ______________________________

**Date:** ____________________ **Period:** ____________________

Change one part of the holder. Keep the parcel, speed, time, floor and start line the same.

## A successful test

- ☐ The holder clears the wheels, sensor and controls.
- ☐ The parcel stays on in at least two of three runs.
- ☐ The team records tests for both holders before choosing.

## Record every result

| Holder / run | Parcel stayed on? | Any rubbing? | What happened? |
|---|---|---|---|
| A / 1 | | | |
| A / 2 | | | |
| A / 3 | | | |
| B / 1 | | | |
| B / 2 | | | |
| B / 3 | | | |

# Change one thing. Try again.

The one thing we changed:

We kept these things the same:

| Holder / run | Parcel stayed on? | Any rubbing? | What happened? |
| --- | --- | --- | --- |
| 1 | | | |
| 2 | | | |
| 3 | | | |

Which holder worked better? Use a result from your notes.

Use a result: our change helped / did not help because...

Before leaving: save your code, stop and power off the robot, return parts, and hand in this module.

# Help your robot read colors

**Name:** ______________________________

**Date:** ____________________ **Period:** ____________________

## Today I can...

Test paper colors and choose a setup that gives clear readings.

## Words to know

Reading: The information a sensor gives you.

Unknown reading: The sensor cannot name the color.

Fair test: Change one thing at a time.

**LOOK AT THE IDEA**


## Gather your materials

- Checked base with its downward-facing sensor
- Thin card about 30 cm by 12 cm
- Red, white and blue paper squares, each 8 cm wide
- Removable tape, ruler and pencil

# Build a color-sensor test board

Build with the power off. Tick each step when it is done.

1. Tape the three paper squares side by side on the card: red, white and blue. Leave a small gap between them. Keep the paper flat, without shiny tape over the reading area.
2. Keep the robot’s wheels raised on firm supports. Slide the board under the sensor. Keep every motor stopped for this lesson.
3. Mark the board position that puts the center of red under the sensor. Do the same for white and blue. These marks help you line up each test.
4. Measure the gap from the sensor face to the board. Ask the teacher to check it against LEGO’s guidance. Record the gap; do not assume one height works for every build.

**Sketch the setup. Label the parts you will check.**

# Finish the setup

5. Use the program below to show the color name. Test red three times, white three times and blue three times. Move the board away and back to its mark between readings.

6. Write down all nine readings, even wrong or unknown ones. Unknown means the sensor cannot name the color.

7. Change only one thing, such as room lighting or the sensor gap. Repeat all nine readings. Keep the paper and its position the same.

8. Choose the setup that gives the most reliable red and white readings. Record the gap and lighting. Keep the same paper for the stopping lesson.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

**Show where your robot starts and how you will stop it.**

# Code it. Predict. Try it.

## Make the program

1. Start with all movement stopped.

2. Repeat 10 times: read the color, show its name, and wait 0.2 seconds.

3. Stop all movement and end.

Coding Canvas is the app where you make the program. Use your teacher-checked settings before a floor run.

Before you run: what do you think will happen?

After one try: what actually happened?

End of class 1: save the code, stop and power off, label your parts, and keep these pages.

# Test it fairly

**Name:** ______________________________

**Date:** ____________________ **Period:** ____________________

Change the light OR sensor height. Keep the other things the same.

## A successful test

- ☐ Nine readings are recorded for each setup.
- ☐ Only one thing changes in the comparison.
- ☐ The team can explain the chosen red-and-white setup.

## Record every result

| Paper / setup | Reading 1 | Reading 2 | Reading 3 |
|---|---|---|---|
| Red | | | |
| White | | | |
| Blue | | | |

First setup: sensor gap (cm) and room lighting:

______________________________

# Change one thing. Try again.

The one thing we changed:

We kept these things the same:

| Paper / setup | Reading 1 | Reading 2 | Reading 3 |
|---|---|---|---|
| Red | | | |
| White | | | |
| Blue | | | |

What did you change, and did it help the robot read colors?

Use a result: our change helped / did not help because...

Before leaving: save your code, stop and power off the robot, return parts, and hand in this module.

# Plan the delivery route

**Name:** ______________________________

**Date:** ____________________ **Period:** ____________________

## Today I can...

Put steps in order, then test and adjust a short route.

## Words to know

Sequence: Steps in order.

Calibrate: Test and adjust the settings.

Variable: One thing you can change, such as drive time.

**LOOK AT THE IDEA**

**1. FORWARD** → **2. RIGHT** → **3. FORWARD** → **4. STOP**

Test each part. Stop between parts. Then join the steps.

## Gather your materials

- Checked base, ruler and removable floor tape
- Six paper cards about 8 cm square
- Pencil or marker
- The short-drive code from lesson 2

# Build a route with arrow cards

Build with the power off. Tick each step when it is done.

1. Draw forward arrows on four cards, a right-turn arrow on one card and STOP on the last card. These cards are a plan; they do not control the robot by themselves.
2. Mark a short straight lane and a wide delivery space on the floor. Leave enough clear room for a turn. Keep arrow cards beside the lane, away from the wheels.
3. Arrange the cards: drive forward, turn right, drive forward, stop. Point to the direction the robot will face after each step.
4. Test the first short drive on its own. Adjust its time until it reaches the turning area. Calibrate means test and adjust a setting.

**Sketch the setup. Label the parts you will check.**

# Finish the setup

5. With the teacher, test a short, slow right turn by itself. Stop after the turn. Change its time a little if it turns too far or not far enough.
6. Test the second short drive. Join the three checked parts, with a Stop after each one. Keep the first complete tests to 3 seconds of movement or less.
7. Run the route three times from the same mark and direction. Record where the robot stops each time. Keep people and objects out of the lane.
8. Change one time setting and try another set of runs. Save the code and draw the final route.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

**Show where your robot starts and how you will stop it.**

# Code it. Predict. Try it.

## Make the program

1. Start stopped. Drive slowly for the tested first time, then stop.

2. Turn right slowly for the tested turn time, then stop.

3. Drive slowly for the tested second time, then stop and end.

Coding Canvas is the app where you make the program. Use your teacher-checked settings before a floor run.

Before you run: what do you think will happen?

After one try: what actually happened?

End of class 1: save the code, stop and power off, label your parts, and keep these pages.

# Test it fairly

**Name:** ______________________________

**Date:** ____________________ **Period:** ____________________

Change one time setting. Keep the speed, floor, load and starting direction the same.

## A successful test

- ☐ The program stops after each part and at the end.
- ☐ The robot finishes inside the space in at least two of three runs.
- ☐ The team records the times and layout so it can try again.

## Record every result

| Run | Drive / turn / drive (seconds) | Inside space? | What happened? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

Which result stands out?

______________________________

# Change one thing. Try again.

The one thing we changed:

We kept these things the same:

| Run | Drive / turn / drive (seconds) | Inside space? | What happened? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

Which time setting did you change, and what happened?

Use a result: our change helped / did not help because...

Before leaving: save your code, stop and power off the robot, return parts, and hand in this module.

# Stop at the red line

**Name:** ______________________________

**Date:** ____________________ **Period:** ____________________

## Today I can...

Make the robot move on white and stop on red or an unknown reading.

## Words to know

If-then rule: If this happens, do that.

Loop: Steps that repeat.

Bounded: Set to stop after a certain time or number of steps.

**LOOK AT THE IDEA**

| WHITE | RED | OTHER |
| --- | --- | --- |
| ↓ | ↓ | ↓ |
| **Move a little** | **Stop** | **Stop and check** |

## Gather your materials

- Checked base with downward-facing color sensor
- The tested white and red paper from lesson 4
- Removable tape and ruler
- Connected device with lesson 6 code

# Build a red-line stopping station

Build with the power off. Tick each step when it is done.

1. Make a flat white lane at least twice as wide as the robot. Tape its edges down so the wheels cannot catch them. Keep tape away from the sensor’s reading path.
2. Place a red strip across the full lane, starting about 10 cm deep in the direction of travel. This is a starting test size, not a guaranteed stopping distance.
3. Leave a large clear area beyond the red strip. Mark a start close enough that a short drive can reach it. Ask the teacher to check the whole layout.
4. With the wheels still, show the sensor white, red and another color. Fix wrong readings before driving. Keep the sensor gap from lesson 4.

**Sketch the setup. Label the parts you will check.**

# Finish the setup

5. Build the program below. A variable is a value the program remembers. keepGoing remembers whether this test may continue.

6. At low speed, try one approach to red. Keep someone ready at Stop. When the program ends, measure how far the sensor moved past the start of the red strip.

7. Repeat three approaches. Separately test an unknown or other-color reading with wheels raised. After a red or unknown reading, show white again: the same test must stay stopped.

8. If it misses red, stop. Try a wider strip or a lower speed, one change at a time. Do not remove the 20-check limit or the Stop steps.

- [ ] Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

**Show where your robot starts and how you will stop it.**

# Code it. Predict. Try it.

## Make the program

1. Start stopped. Set keepGoing to Yes.

2. Repeat at most 20 times. Only if keepGoing is Yes: read the color.

3. If white: move slowly for 0.1 seconds, then stop. Otherwise: stop and set keepGoing to No.

4. After the repeats, stop and end. A person must start a new test.

Coding Canvas is the app where you make the program. Use your teacher-checked settings before a floor run.

keepGoing is a value the program remembers: Yes allows another check; No keeps the robot stopped.

Before you run: what do you think will happen?

After one try: what actually happened?

End of class 1: save the code, stop and power off, label your parts, and keep these pages.

# Test it fairly

**Name:** ______________________________

**Date:** ______________________ **Period:** ______________________

Change speed OR strip width. Keep the light, sensor height, start and white paper the same.

## A successful test

- ☐ Three approach runs stop when red is read.
- ☐ Red and unknown both end the test; white cannot restart it.
- ☐ The program ends after no more than 20 checks.

## Record every result

| Run | Saw red? | Distance past line (cm) | Stayed stopped? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

- ☐ Wheels raised: red and other/unknown readings stop it. Showing white again does not restart this test.

# Change one thing. Try again.

The one thing we changed:

We kept these things the same:

| Run | Saw red? | Distance past line (cm) | Stayed stopped? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

What could make the robot miss a narrow red strip?

Use a result: our change helped / did not help because...

Before leaving: save your code, stop and power off the robot, return parts, and hand in this module.

# Make the delivery

**Name:** ______________________________

**Date:** ____________________ **Period:** ____________________

## Today I can...

Join your tested parts and check the whole delivery three times.

## Words to know

Success criteria: The things that must happen for a test to count as a success.

Version: One saved design or program before you change it.

Reliable: Works as expected again and again.

**LOOK AT THE IDEA**


## Gather your materials

- Checked base and the selected holder from lesson 3
- The same light paper parcel
- White lane and red strip from lesson 6
- Removable tape, ruler and saved code

# Build the complete delivery course

Build with the power off. Tick each step when it is done.

1. Turn the robot off. Attach the tested holder to the approved mounting points. Put the paper parcel low in the center. Check that no part touches a wheel or covers the sensor.
2. Keep the white approach and red strip from lesson 6. Tape a delivery rectangle around the stopping area, large enough for the whole robot.
3. Keep the lane clear. For this test, use a short straight approach to the red strip. Add a longer route only after this works.
4. Use the checked color-stop program from lesson 6. Keep its short moves, Stop steps and maximum of 20 checks.

**Sketch the setup. Label the parts you will check.**

# Finish the setup

5. Write three rules before driving: parcel stays on; robot stops inside the delivery space; robot hits nothing. A run passes only when all three happen.

6. Run the delivery three times from the same start. Write Yes or No for each rule after each run. Keep the failed runs in the notes.

7. Choose one change, such as a lower holder wall or wider red strip. Keep the other parts and code the same. Try another set of three runs.

8. Save the best-tested setup. Keep the base ready for the final build choice.

- [ ] Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

**Show where your robot starts and how you will stop it.**

# Code it. Predict. Try it.

## Make the program

1. Use the complete lesson 6 color-stop program.

2. White permits one 0.1-second slow move and then Stop. Red or any other reading ends the test.

3. Stop and end after at most 20 checks. Never restart within the same test.

Coding Canvas is the app where you make the program. Use your teacher-checked settings before a floor run.

keepGoing is a value the program remembers: Yes allows another check; No keeps the robot stopped.

Before you run: what do you think will happen?

After one try: what actually happened?

End of class 1: save the code, stop and power off, label your parts, and keep these pages.

# Test it fairly

**Name:** ______________________________

**Date:** ______________________ **Period:** ______________________

Change one thing between versions. Keep everything the same during each set of three runs.

## A successful test

- ☐ At least two of three runs meet all three rules.
- ☐ The robot still passes the red-and-unknown stop checks.
- ☐ The team can name one change and the result it produced.

## Record every result

| Run | Parcel stayed on? | Inside space? | Hit nothing? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

Which result stands out?

______________________________

# Change one thing. Try again.

The one thing we changed:

We kept these things the same:

| Run | Parcel stayed on? | Inside space? | Hit nothing? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

What improved when you changed one part of the delivery?

Use a result: our change helped / did not help because...

Before leaving: save your code, stop and power off the robot, return parts, and hand in this module.

# Choose your robot's next job

**Name:** ____________________

**Date:** ____________________ **Period:** ____________________

Choose one of five builds, test it and explain how you improved it.

## Choose ONE build

### 1. Parcel delivery robot - full module: 5-8

Carry a paper parcel to a red-line delivery space.

### 2. Drawing robot - full module: 9-12

Draw a short line on a protected paper surface.

### 3. Paper sweeper - full module: 13-16

Push three light paper pieces into a collection area.

### 4. Ball-pushing robot - full module: 17-20

Guide one light paper ball toward a goal.

### 5. Rescue carrier - full module: 21-24

Tow a light paper figure on a small rescue sled.

Your teacher will give you the pages for one build. Class 1: choose and build. Class 2: test, improve and share. The page numbers above refer to the complete module workbook.

# Plan before you build

Our chosen job:

Why we chose it:

**LOOK AT THE IDEA**

**BUILD** → **TEST** → **IMPROVE**

Keep your notes. Change one thing. Try again.

## Words to know

Attachment: A part you add to the robot for a new job.

Evidence: Measurements and notes that show what happened.

Handover: Directions that help someone else use your robot.

**Draw the attachment and label where it joins the base.**

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

# Test and prepare to share

**Name:** ______________________

**Date:** ______________________ **Period:** ______________________

Use your choice pages to record three runs, change one thing, and test three more times. Keep results from runs that did not work.

## Write your handover directions

Set up the test area:

Start the program:

Stop the robot:

Return to the start safely:

# Show what you learned

Give a two-minute demonstration. If it fails, stop safely, explain what happened and show your earlier results.

Our robot's job and how the code works:

A change we made, with a result from before and after:

My part in the team's work:

Another team's feedback and one next test:

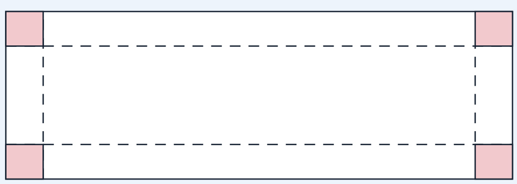
# Choice 1: Parcel delivery robot

**Name:** ____________________

**Date:** ____________________ **Period:** ____________________

Carry a paper parcel to a red-line delivery space.

**LOOK AT THE IDEA**



Cut out the four shaded corners.

Fold on the dashed lines.

Tape corners to make a low tray.

## Gather your materials

- The shared checked base
- Thin card about 12 by 8 cm
- One paper parcel about 4 by 3 by 2 cm
- Ruler, scissors and removable tape

Use the same teacher-checked base. The drawing explains the attachment; it is not a full-size cutting template.

# Build your attachment

1. Measure the clear mounting space on the base. Cut the card to fit it. Keep at least 1 cm of room between the attachment and any wheel. Make it smaller if needed.
2. Draw a fold line 1 cm inside each edge. Cut out the four corner squares. Fold all four walls up and tape the corners to make a shallow tray.
3. Fold a light paper parcel. Place it in the center of the tray. Add a small folded-paper stop at each end if it slides. Do not add heavy contents.
4. Turn the robot off. Use the teacher-approved mounting points and removable connectors to hold the tray flat. Keep buttons and the sensor clear.
5. Raise the wheels and check for rubbing. Then place the robot on the white lane from lesson 6. Keep the broad red strip and empty space beyond it.
6. Run the code below three times. If the parcel falls, change one tray feature and repeat. Unload the parcel by hand only after the robot has stopped.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

# Program and first tests

**Name:** ______________________________

**Date:** ____________________ **Period:** ____________________

1. Use lesson 6’s tested color-stop program without removing its limits.

2. Start stopped; set keepGoing to Yes. Repeat at most 20 times. Only while keepGoing is Yes, read the color: white means a slow 0.1-second move then Stop; any other reading means Stop and set keepGoing to No.

3. Stop and end after the repeats. Red means delivery; unknown means check the setup.

keepGoing remembers whether this test may continue. Yes allows a check; No keeps the robot stopped.

My prediction:

| Run | Parcel stayed on? | Inside space? | Hit nothing? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

# Improve your chosen build

## What counts as success?

- ☐ The parcel stays in the tray.
- ☐ The whole robot stops inside the delivery space.
- ☐ The robot hits no object or person.

Change one thing. What did you change?

| Run | Parcel stayed on? | Inside space? | Hit nothing? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

Which tray feature helped the parcel stay on?

Which result supports your answer?

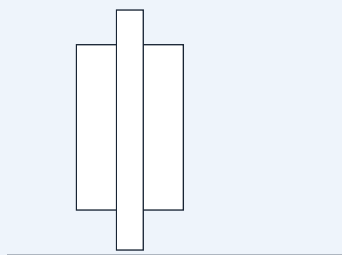
# Choice 2: Drawing robot

**Name:** ______________________

**Date:** ______________ **Period:** ______________

Draw a short line on a protected paper surface.

LOOK AT THE IDEA



**Card sleeve**

Marker tip just touches the paper.

Wheels stay on the surface.

## Gather your materials

- The shared checked base
- One washable felt-tip marker
- A card strip about 12 by 3 cm and removable tape
- A large sheet of paper over a protective mat

Use the same teacher-checked base. The drawing explains the attachment; it is not a full-size cutting template.

# Build your attachment

1. Cover a large floor area with a protective mat and paper. Tape the paper edges flat. Ask the teacher to approve the marker and surface.
2. Wrap the card strip loosely around the marker to make a sleeve. Tape the card to itself. The marker must slide up and down inside it.
3. Turn the robot off. Fix the sleeve upright at a teacher-approved point behind the wheels. Keep it away from the sensor and moving parts.
4. Slide the uncapped marker down until its tip just touches the paper. Secure the marker in the sleeve with a small removable tape tab. It must not lift the wheels off the paper.
5. Raise the wheels and check the short program. Put the robot back on the protected paper with enough space for the whole run.
6. Draw three short lines. Stop and turn off the robot before lifting or adjusting the marker. If a line has gaps, lower the tip a little; if the robot struggles, raise it.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

# Program and first tests

**Name:** ____________________

**Date:** ____________________ **Period:** ____________________

1. Start with movement stopped. Use a checked low speed, such as 20 percent.

2. Drive forward for 0.8 seconds, then stop and end. Shorten the time if the paper is too small.

3. No automatic repeat. Lift the marker and return the robot by hand between runs with the power off.

My prediction:

| Run | Visible line? | Stayed on paper? | Stopped by itself? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

# Improve your chosen build

## What counts as success?

☐ A visible line is drawn on the paper.

☐ The marker stays in its holder and all marks stay on protected paper.

☐ The robot stops by itself at the programmed time.

Change one thing. What did you change?

| Run | Visible line? | Stayed on paper? | Stopped by itself? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

How did marker height change the line or the robot’s movement?

Which result supports your answer?

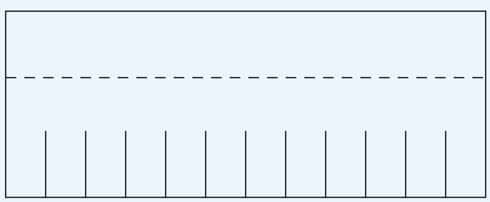
# Choice 3: Paper sweeper

**Name:** ______________________________

**Date:** ____________________ **Period:** ____________________

Push three light paper pieces into a collection area.

**LOOK AT THE IDEA**



Fold the top edge back.

Cut soft bristles below the fold.

Keep the wheels clear.

## Gather your materials

- The shared checked base
- A thin card strip about 12 by 5 cm
- Three tissue-paper squares, each about 2 cm wide
- Removable tape, ruler and scissors

Use the same teacher-checked base. The drawing explains the attachment; it is not a full-size cutting template.

# Build your attachment

1. Cut a card strip as wide as the robot, but no wider than the clear test lane. Fold its top 2 cm backward to make a mounting edge.
2. Cut short vertical slits about 1 cm apart into the lower 2 cm. These make flexible paper bristles. Round sharp card corners with scissors.
3. Turn the robot off. Attach the folded top edge to approved fixed front mounting points. The flexible ends should just brush the floor without lifting any wheel.
4. Use a clear floor lane. Mark a collection box farther along it. Place three tissue squares in a row across the starting sweep area. Keep them away from wheel paths.
5. Try the short program below. Check that the paper bristles can bend. Stop at once if a piece catches in a wheel; turn the power off before removing it.
6. Count the pieces that reach the box. Repeat three times with the same starting positions. Adjust the bristle height once and compare.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

# Program and first tests

**Name:** ______________________________

**Date:** ____________________ **Period:** ____________________

1. Start stopped and set the checked low speed.

2. Drive forward for 0.8 seconds, then stop and end. Set the collection box using the distance measured in lesson 2.

3. Do not add continuous driving. Return the paper pieces and robot to their marks between tests.

My prediction:

______________________________

| Run | Pieces in box (0-3) | Wheels clear? | Stopped by itself? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

# Improve your chosen build

## What counts as success?

- ☐ At least two of the three tissue squares reach the box.
- ☐ No paper gets caught in the wheels.
- ☐ The robot stops by itself and touches no object except the tissue squares.

Change one thing. What did you change?

| Run | Pieces in box (0-3) | Wheels clear? | Stopped by itself? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

Did shorter or longer bristles move the paper better?

Which result supports your answer?

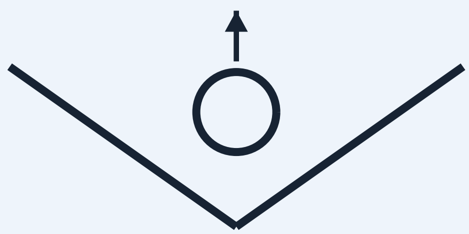
# Choice 4: Ball-pushing robot

**Name:** ____________________

**Date:** ____________________ **Period:** ____________________

Guide one light paper ball toward a goal.

**LOOK AT THE IDEA**



**Top view: a shallow V**

The opening faces forward.

Leave room for the ball to roll.

## Gather your materials

- The shared checked base
- Thin card about 14 by 5 cm
- One crumpled paper ball about 3 cm wide
- Removable tape and ruler

Use the same teacher-checked base. The drawing explains the attachment; it is not a full-size cutting template.

# Build your attachment

1. Fold the card lengthwise so it has a 3 cm upright wall and a 2 cm mounting flap. Fold it gently across the middle to make a shallow V.

2. Make one light paper ball about 3 cm wide. Put it in the V opening. Leave enough room for it to roll freely; do not make a tight clamp.

3. Turn the robot off. Attach the two mounting flaps to teacher-approved fixed front points. Keep the V open at the front. Leave the wheels clear and keep the card just above the floor.

4. Keep the sensor clear even though this task uses a timed drive. Mark a wide goal across a short straight lane. Place the ball at the same start mark each time.

5. Use the short program below. The robot pushes the ball; it does not grip it. Stop if the ball rolls toward people or outside the clear test area.

6. Repeat three times. If the ball escapes sideways, adjust the V angle a little. Keep the same ball and code while comparing.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

# Program and first tests

**Name:** ____________________

**Date:** ____________________ **Period:** ____________________

1. Start stopped and use the checked low speed.

2. Drive forward for 0.8 seconds, then stop and end. Put the goal within the tested travel distance.

3. No automatic repeat. Return the robot and ball to their marks with the power off.

My prediction:

| Run | Ball crossed goal? | Guide stayed on? | Stopped by itself? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

# Improve your chosen build

## What counts as success?

- ☐ The ball crosses the marked goal line.
- ☐ The guide stays attached and the wheels turn freely.
- ☐ The robot stops by itself and touches only the paper ball.

Change one thing. What did you change?

| Run | Ball crossed goal? | Guide stayed on? | Stopped by itself? |
| --- | --- | --- | --- |
| 1 | | | |
| 2 | | | |
| 3 | | | |

Which V angle kept the ball in front of the robot?

Which result supports your answer?

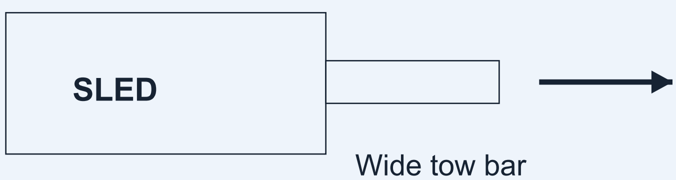
# Choice 5: Rescue carrier

**Name:** ______________________________

**Date:** ____________________ **Period:** ____________________

Tow a light paper figure on a small rescue sled.

**LOOK AT THE IDEA**




Attach behind the robot.

Keep away from wheels.

Use a straight route.

## Gather your materials

- The shared checked base
- Thin card: one 12 by 6 cm piece and one 12 by 2 cm strip
- A flat paper person, two paper strips for bands, removable tape and scissors
- Ruler and a clear straight lane

Use the same teacher-checked base. The drawing explains the attachment; it is not a full-size cutting template.

# Build your attachment

1. Use the 12 by 6 cm card as a sled. Fold each long edge up by 1 cm. Curl the front edge slightly upward so it does not catch on the floor.
2. Draw and cut out a small paper person. Lay it flat on the sled. Add two loose paper bands across the sled and tape their ends to the sides to hold the figure in place.
3. Use the 12 by 2 cm card strip as a wide tow bar. Tape one end flat to the sled’s front. Make a small folded tab at the other end.
4. Before attaching it to the robot, pull the sled gently by hand using its wide tow bar. It should slide without catching. Keep it light and use only a straight route for this task.
5. Turn the robot off. Ask the teacher to connect the tow-bar tab to a fixed rear mounting point. Keep the whole strip centered behind the robot and away from the wheels. Do not use loose string.
6. Run the short program below three times. If the sled catches or a wheel struggles, stop. Smooth the sled edge or shorten the route, then test again.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

# Program and first tests

**Name:** ______________________________

**Date:** ____________________ **Period:** ____________________

1. Start stopped and use the checked low speed.
2. Drive straight for 0.8 seconds, then stop and end. Do not add turns or reversing for this first sled build.
3. Return the robot and sled to the start with the power off. Do not move a real person, animal or heavy object.

My prediction:

______________________________

| Run | Person stayed on? | Tow bar clear? | Moved and stopped? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

# Improve your chosen build

## What counts as success?

- ☐ The paper person stays on the sled.
- ☐ The tow bar stays attached and away from the wheels.
- ☐ The robot moves the sled and stops by itself.

Change one thing. What did you change?

| Run | Person stayed on? | Tow bar clear? | Moved and stopped? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

What helped the sled slide without catching?

Which result supports your answer?

# Word help

### Rover
A small robot that moves on wheels.

### Program
A list of steps a computer or robot follows.

### Coding block
A piece on the screen that gives the robot an instruction.

### Sensor
A part that detects something, such as a color.

### Motor
A part that uses electricity to make something move.

### Input / output
Input is information going in. Output is the action or message that comes out.

### Bounded
Set to stop after a certain time or number of steps.

### Trial
One test run.

### Fair test
Change one thing. Keep the other things the same.

# Word help

### Cargo / cradle

Cargo is what the robot carries. A cradle is a holder that keeps it in place.

### Calibrate

Test and adjust settings so the robot does what you need.

### Sequence / loop

A sequence is steps in order. A loop repeats steps.

### Variable

One thing that can change, such as speed or time.

### Mean

An average: add the numbers, then divide by how many numbers you added.

### Success criteria

The things that must happen for a test to count as a success.

### Evidence

Measurements or notes that show what happened.

### Attachment

A part you add to the robot for a new job.

# Kit cost planning

US planning prices checked September 10, 2026. Set #45522. Check current prices before buying.

| What to buy | USD | What this means |
|---|---|---|
| One kit · up to 4 learners | $529.95 | LEGO’s listed price |
| Six individual kits · 24 learners | $3,179.70 | 6 × $529.95; fewer charging extras than bundle |
| Classroom bundle · 24 learners | $3,499.00 | 6 kits plus charging and replacement extras |

The kit supports a team of four. The class bundle has six kits, three extra multi-chargers, eight extra connection cards and replacement bricks.

View the official LEGO kit and current price

# Kit cost planning

US planning prices checked September 10, 2026. Set #45522. Check current prices before buying.

| What to buy | USD | What this means |
|---|---|---|
| Supplies for a small tryout | $35.00 | Money set aside for: paper, tape, ruler and cargo materials |
| Supplies for six teams | $120.00 | Money set aside for six teams |
| Small tryout total before tax / shipping | $564.95 | 1 kit + $35 supplies; using a device you already have |
| Class total before tax / shipping | $3,619.00 | 1 bundle + $120 supplies; using 6 devices you already have |

Prices were checked on LEGO’s US website. Check them again before buying. The totals do not include tax, shipping, computers, school computer support or teacher training. The course and screen practice area do not need a paid subscription. No kit has been bought.

View the official LEGO kit and current price